

Home Server Handbook

The home server is a refurbished Dell OptiPlex 7050 with an Intel i5-7500, 16 GB of RAM and two 4 TB hard drives in a ZFS mirror called tank. It runs Ubuntu Server 24.04 LTS and sits in the hallway cupboard next to the router.

Every service runs as a Docker container managed with Docker Compose; the compose files live in `/srv/compose`, one folder per service. The main services are Jellyfin for films and music, Nextcloud for files and calendars, and Home Assistant for the smart plugs and the thermostat.

This handbook is for whoever looks after the server. Keep it up to date when something changes.

Backups

Nightly backups run at 02:00 with restic. Snapshots go to the external USB drive labelled BACKUP-A, and every Sunday they are copied to a Backblaze B2 bucket as the off-site copy.

Seven daily, four weekly and six monthly snapshots are kept. Old snapshots are pruned after the Sunday copy has finished.

At 01:30, before the backup starts, the Nextcloud PostgreSQL database is dumped to /srv/dumps, so every snapshot holds a consistent copy of it.

The backup password is stored in the family password manager under "restic repository". Without it the backups cannot be read, so never change it without updating the entry.

Restoring files

To get files back, list the snapshots with `restic snapshots`, then run `restic restore` with the snapshot ID and `--target /tmp/restore`. Copy the files you need back yourself rather than restoring over the live folders.

Test a restore on the first Saturday of every month, and note the result in the log at the end of this handbook.

If the whole server has to be rebuilt, install Ubuntu Server on the new disk, import the ZFS pool with `zpool import tank`, reinstall Docker, then run `docker compose up -d` in each folder under `/srv/compose`.

Updates and monitoring

Security updates install automatically with unattended-upgrades. If an update needs a restart, the server reboots itself at 04:00 on Wednesdays.

Container images are updated by hand once a month: docker compose pull, then docker compose up -d, one service at a time, checking each one still works.

Uptime Kuma checks each service every five minutes and posts in the family Signal group when one stops responding.

A ZFS scrub runs on the second Sunday of each month, and a SMART report on both drives is emailed every Monday.